

Adopted Levels, Gammas

$Q(\beta^-)=3988.4$ 19; $S(n)=8380.4$ 20; $S(p)=12155.1$ 20; $Q(\alpha)=-12332.0$ 29 2021Wa16

$S(2n)=14663.1$ 22, $S(2p)=30938$ 7 (2021Wa16).

Isotope discovery (2012Th10): $^{232}\text{Th}(^{40}\text{Ar},X)$ at Dubna (1971Ar32) and $^{37}\text{Cl}(\gamma,2p)^{35}\text{P}$ at Mainz (1971Gr53).

^{35}P production:

2012Kw02: $E(^{40}\text{Ar})=140$ MeV/nucleon impinged on ^9Be and natural nickel targets at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

2012Zh06: $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=57$ MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.

2007No13: $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$ and $(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=100$ MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.

1997Vo03: $^{56}\text{Fe}(p,X)$ at $E_p=800$ MeV at LANL. Measured γ radiation. Deduced production cross sections.

^{35}P decay measurements:

1986Wa22: ^{35}P produced via $^{36}\text{S}(t,\alpha)^{35}\text{P}$ at BNL. Measured β -delayed γ rays.

1972Ap01: ^{35}P produced via $^{37}\text{Cl}(t,\alpha p)^{35}\text{P}$ at LANL. Measured $T_{1/2}$ and β -delayed γ rays.

1972Go31: ^{35}P produced via $^{18}\text{O}(^{19}\text{F},2p)^{35}\text{P}$ and $^{36}\text{S}(t,\alpha)^{35}\text{P}$ at BNL. Measured $T_{1/2}$ and β -delayed γ rays.

1971Gr53: ^{35}P produced via $^{37}\text{Cl}(\gamma,2p)^{35}\text{P}$ at Mainz. Measured $T_{1/2}$ and γ rays.

^{35}P radius measurement:

1999Ai02: $\text{Si}(^{35}\text{P},X)$ at NSCL. Measured energy-integrated reaction cross sections at $E=38$ -80 MeV/ nucleon. Deduced strong absorption radii.

^{35}P mass measurements:

$^{34}\text{S}(^{18}\text{O},^{17}\text{F})$ and $^{37}\text{Cl}(^{11}\text{B},^{13}\text{N})$ (1988Or01), $^{36}\text{S}(^6\text{Li},^7\text{Be})$ (1985Dr06), $^{36}\text{S}(d,^3\text{He})$ (1985Kh04), $^{36}\text{S}(^{14}\text{C},^{15}\text{N})$ (1984Ma49).

Theoretical calculations (binding energies, dipole moments, quadrupole moments, radii, levels, J , π , etc.): 2012BoZT, 2009No01, 2004Kh16, 2003Sm02, 1999Du05, 1988Wa04, 1987Wa10, 1986Wo02, 1983Wi08, 1975JeZX.

 ^{35}P LevelsCross Reference (XREF) Flags

A	$^{35}\text{Si} \beta^-$ decay (0.78 s)	E	$^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$	I	$^{37}\text{Cl}(^{11}\text{B}, ^{13}\text{N})$
B	$^1\text{H}(^{34}\text{Si}, p): \text{resonances}$	F	$^{34}\text{S}(^{18}\text{O}, ^{17}\text{F})$	J	$^{160}\text{Gd}(^{37}\text{Cl}, X\gamma)$
C	$^2\text{H}(^{34}\text{Si}, ^{35}\text{P}\gamma)$	G	$^{36}\text{S}(\text{pol } d, ^3\text{He})$	K	$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$
D	$^2\text{H}(^{36}\text{S}, ^3\text{He})$	H	$^{36}\text{S}(d, ^3\text{He})$		

$E(\text{level})^\dagger$	J^π	$T_{1/2}$ or $\Gamma^\#$	XREF	Comments
0	$1/2^+$	47.3 s 8	A CDEFGHIJK	$\% \beta^- = 100$ J^π : L(pol d, ^3He)=0 from 0^+ and analyzing power. $T_{1/2}$: weighted average of 45 s 2 (1971Gr53, $\gamma(t)$), 47.7 s 29 (1972Go31, $\gamma(t)$), 48.2 s 17 (1972Go31, $\gamma(t)$), and 47.4 s 8 (1972Ap01, $\gamma(t)$). XREF: F(2420)
2386.9 11	$3/2^+$	<0.69 ps	A CDEFG I K	J^π : L(pol d, ^3He)=2 from 0^+ and L-1/2 transfer from analyzing power.
3860.4 11	$5/2^+$	<0.69 ps	A CDE GHIJK	J^π : L(pol d, ^3He)=2 from 0^+ and L+1/2 transfer from analyzing power.
4101.7 11	$(7/2^-)^\ddagger$	>69 ps	A C E JK	
4250 20			I	
4382.0 12	$(5/2^-)$		A E K	XREF: A(?) J^π : possibly allowed β^- feeding from $7/2^-$ parent; 1994.9 γ to $3/2^+$.
4494.1 12	$(7/2^-)^\ddagger$	2.29 ps 49	A C E G JK	XREF: G(4474)
4666.2 16	$5/2^+$		DE GHI	J^π : L=(3) in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$ from 0^+ . XREF: I(4640)
4767.0 13	$(9/2^-)^\ddagger$		E K	J^π : L(pol d, ^3He)=2 from 0^+ and L+1/2 transfer from analyzing power.

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}P Levels (continued)				
E(level) [†]	J ^π	T _{1/2} or Γ [#]	XREF	Comments
4869.6 12	(5/2 ⁻ , 7/2 ⁻)		A C K	J ^π : possibly allowed β ⁻ feeding from 7/2 ⁻ parent; 1009.7γ to 5/2 ⁺ .
4962.8 12	(9/2 ⁻) [‡]		A E K	XREF: A(?)
5010 20			I	1988Or01 observed a 5010 20 level in ($^{11}\text{B}, ^{13}\text{N}$) and a 5070 40 level in ($^{18}\text{O}, ^{17}\text{F}$) but did not assign them as the same level.
5090.2 13	(11/2 ⁻) [‡]		EF K	XREF: F(5070)
5199.3 16	5/2 ⁺		DE GHI	XREF: I(5220) J ^π : L(pol d, ^3He)=2 from 0 ⁺ and L+1/2 transfer from analyzing power.
5487.9 13			K	
5561.0 13	(5/2 ⁻)		A K	J ^π : possibly allowed β ⁻ feeding from 7/2 ⁻ parent; 3173.5γ to 3/2 ⁺ .
5709.5 23	(1/2 ⁻)		DE	J ^π : L=(1) in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$ from 0 ⁺ ; interpreted as the deeply bound 1p _{1/2} proton removal from 0 ⁺ in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$; 5709γ to 1/2 ⁺ .
5.86×10 ³ 5			F I	XREF: F(5890)I(5840) E(level): weighted average of 5890 70 from ($^{18}\text{O}, ^{17}\text{F}$) and 5840 50 from ($^{11}\text{B}, ^{13}\text{N}$).
6222.7 13	(7/2 ⁻ , 9/2, 11/2 ⁻)		K	J ^π : 1132γ to (11/2 ⁻) and 1729γ to (7/2 ⁻).
6440 60			F	
7050 60			F	
7440 60			Fg	XREF: g(7520)
7526.9 23	(1/2 ⁻)		E g	XREF: g(7520) J ^π : L=(1) in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$ from 0 ⁺ ; interpreted as the deeply bound 1p _{1/2} proton removal from 0 ⁺ in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$; 7526γ to 1/2 ⁺ .
7590 20			I	
7920 60			F	
8390 40			I	
8.60×10 ³ 10			F	
9290 50			F	
14938 24	1/2 ⁺	<12.7 keV	B	J ^π : L=0 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
15161 3	5/2 ⁻ , 7/2 ⁻	<4.4 keV	B	J ^π : L=3 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
15306 24	3/2 ⁺ , 5/2 ⁺	<30.4 keV	B	J ^π : L=2 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
15964 18	3/2 ⁺ , 5/2 ⁺	84 keV 25	B	J ^π : L=2 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
16145 36	1/2 ⁻ , 3/2 ⁻	0.35 MeV 9	B	J ^π : L=1 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
16605 44	1/2 ⁺	0.22 MeV 15	B	J ^π : L=0 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
17254 12	3/2 ⁺ , 5/2 ⁺	<11.6 keV	B	J ^π : L=2 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
17355 15	1/2 ⁻ , 3/2 ⁻	32 keV 22	B	J ^π : L=1 in $^1\text{H}(^{34}\text{Si}, \text{p})$.

[†] From a least-squares fit to γ-ray energies for levels connected with γ transitions, from particle-transfer reactions for other levels, or from proton elastic scattering for resonances.

[‡] Comparisons with shell-model calculations (2019Gr08).

[#] T_{1/2} from the differential recoil-distance method (2019Gr08) in $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ and widths from the R-matrix analysis of ($^{34}\text{Si}, \text{p}$) for resonances, unless otherwise noted.

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{P})$									
$E_i(\text{level})$	J_i^π	E_γ	I_γ	E_f	J_f^π	Mult.	δ	$\alpha^\#$	Comments
2386.9	3/2 ⁺	2386.3 6	100	0	1/2 ⁺	[M1,E2]		0.00046 5	E_γ : weighted average of 2386.4 6 from $^{35}\text{Si } \beta^-$ decay, 2386 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 2386 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. B(M1)(W.u.)>0.0023 if M1, B(E2)(W.u.)>1.6 if E2.
3860.4	5/2 ⁺	1473.5 5	15.6 14	2386.9	3/2 ⁺	[M1,E2]		8.3×10 ⁻⁵ 13	E_γ : weighted average of 1473.4 5 from $^{35}\text{Si } \beta^-$ decay, 1473 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 1474 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : weighted average of 14.1 33 from $^{35}\text{Si } \beta^-$ decay and 15.9 14 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. B(M1)(W.u.)>0.0012 if M1, B(E2)(W.u.)>2.1 if E2.
		3860.2 10	100.0 32	0	1/2 ⁺	[E2]		1.12×10 ⁻³ 2	B(E2)(W.u.)>0.12 E_γ : weighted average of 3859.5 10 from $^{35}\text{Si } \beta^-$ decay, 3860 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 3861 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : From $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. Other: 100 7 from $^{35}\text{Si } \beta^-$ decay.
4101.7	(7/2 ⁻)	241.3 5	100 [†] 7	3860.4	5/2 ⁺	[E1]		0.000665 10	B(E1)(W.u.)<4.4×10 ⁻⁴ E_γ : weighted average of 241.4 3 from $^{35}\text{Si } \beta^-$ decay, 237 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 241 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : other: 100 4 from $^{35}\text{Si } \beta^-$ decay.
		1714.8 6	6.6 [†] 17	2386.9	3/2 ⁺	[M2]		7.93×10 ⁻⁵ 11	B(M2)(W.u.)<0.16 E_γ : weighted average of 1714.7 6 from $^{35}\text{Si } \beta^-$ decay and 1715 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : other: 22 5 from $^{35}\text{Si } \beta^-$ decay.
		4101.4 10	54 [†] 8	0	1/2 ⁺	[E3]		0.000924 13	B(E3)(W.u.)<4.8 E_γ : weighted average of 4100.8 10 from $^{35}\text{Si } \beta^-$ decay and 4102 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : other: 135 8 from $^{35}\text{Si } \beta^-$ decay.
4382.0	(5/2 ⁻)	1994.9 6	100	2386.9	3/2 ⁺				E_γ : weighted average of 1994.8 6 from $^{35}\text{Si } \beta^-$ decay, 1995 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 1995 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. Placement by 1988DuZS, 2008Wi09, and 2016Mu03. 1988DuZT and 1987Wa10 placed this γ as the 6096 to 4101 transition. 1988Or01 placed this γ as the 6488 to 4493 transition.
4494.1	(7/2 ⁻)	392.3 3	100 5	4101.7	(7/2 ⁻)	[M1+E2]	<0.22	0.000199 12	B(M1)(W.u.)=0.117 +42-29 E_γ : weighted average of 392.3 3 from $^{35}\text{Si } \beta^-$ decay, 391

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{P})$ (continued)

$E_i(\text{level})$	J_i^π	E_γ	I_γ	E_f	J_f^π	Mult.	$\alpha^\#$	Comments
								2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 392 <i>I</i> from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : From $^{35}\text{Si} \beta^-$ decay. Other: 100 <i>I</i> 7 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. δ : deduced by evaluators from RUL=100 for B(E2)(W.u.). B(E1)(W.u.)=2.8×10 ⁻⁴ +8-6 E_γ : weighted average of 633.7 <i>I</i> 5 from $^{35}\text{Si} \beta^-$ decay, 634 <i>I</i> 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 633 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : weighted average of 38 <i>I</i> 5 from $^{35}\text{Si} \beta^-$ decay and 27 <i>I</i> 7 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$.
4494.1	(7/2 ⁻)	633.6 <i>I</i> 5	34 <i>I</i> 5	3860.4	5/2 ⁺	[E1]	4.64×10 ⁻⁵ 7	
4666.2	5/2 ⁺	804 [‡] <i>I</i> 2 2279 [‡] <i>I</i> 2 4668 [‡] <i>I</i> 2		3860.4	5/2 ⁺ 2386.9 3/2 ⁺ 0 1/2 ⁺			
4767.0	(9/2 ⁻)	273 <i>I</i> 1	40.0 [†] <i>I</i> 25	4494.1	(7/2 ⁻)			E_γ : weighted average of 274 <i>I</i> 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ and 273 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$.
		664 <i>I</i> 1	100 [†] <i>I</i> 47	4101.7	(7/2 ⁻)			E_γ : weighted average of 666 <i>I</i> 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ and 664 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$.
4869.6	(5/2 ⁻ , 7/2 ⁻)	374 [†] <i>I</i> 1 487 [†] <i>I</i> 1 767.9 <i>I</i> 4	60 [†] <i>I</i> 20 <40 [†] 100 [†] <i>I</i> 20	4494.1	(7/2 ⁻) (5/2 ⁻) (7/2 ⁻)			E_γ : weighted average of 768.0 <i>I</i> 4 from $^{35}\text{Si} \beta^-$ decay and 767 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : other: 100 <i>I</i> 8 from $^{35}\text{Si} \beta^-$ decay.
		1009.7 <i>I</i> 5	<20 [†]	3860.4	5/2 ⁺			E_γ : weighted average of 1009.9 <i>I</i> 5 from $^{35}\text{Si} \beta^-$ decay and 1009 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : other: 152 <i>I</i> 32 from $^{35}\text{Si} \beta^-$ decay.
4962.8	(9/2 ⁻)	468.9 <i>I</i> 4	100 [†] <i>I</i> 8	4494.1	(7/2 ⁻)			E_γ : weighted average of 468.9 <i>I</i> 4 from $^{35}\text{Si} \beta^-$ decay, 469 <i>I</i> 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 468 <i>I</i> 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$.
		859 [†] <i>I</i> 3	66 [†] <i>I</i> 9	4101.7	(7/2 ⁻)			
5090.2	(11/2 ⁻)	128 <i>I</i> 1	50 [†] <i>I</i> 25	4962.8	(9/2 ⁻)			E_γ : weighted average of 127 <i>I</i> 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ and 128 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$.
		322 <i>I</i> 1	100 [†] <i>I</i> 35	4767.0	(9/2 ⁻)			E_γ : weighted average of 321 <i>I</i> 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ and 322 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$.
5199.3	5/2 ⁺	1337 [‡] <i>I</i> 2 2811 [‡] <i>I</i> 2 5202 [‡] <i>I</i> 2		3860.4	5/2 ⁺ 2386.9 3/2 ⁺ 0 1/2 ⁺			
5487.9		993 [†] <i>I</i> 1 1387 [†] <i>I</i> 1	100 [†] <i>I</i> 20 60 [†] <i>I</i> 20	4494.1	(7/2 ⁻) (7/2 ⁻)			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{P})$ (continued)

<u>E_i(level)</u>	<u>J_i^{π}</u>	<u>E_{γ}</u>	<u>I_{γ}</u>	<u>E_f</u>	<u>J_f^{π}</u>	<u>Comments</u>
5561.0	(5/2 ⁻)	1459.4 7	34 12	4101.7	(7/2 ⁻)	E _{γ} : weighted average of 1459.7 5 from ³⁵ Si β^- decay and 1458 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ P γ). I _{γ} : From ³⁵ Si β^- decay.
		3173.5 10	100 17	2386.9	3/2 ⁺	3173.5 γ is not observed in ²⁰⁸ Pb(³⁶ S, ³⁵ P γ), but the weaker 1459 γ deexciting the same level is observed in ²⁰⁸ Pb(³⁶ S, ³⁵ P γ). Further experiments are needed to resolve the discrepancy. E _{γ} ,I _{γ} : From ³⁵ Si β^- decay.
5709.5	(1/2 ⁻)	5709 [‡] 2		0	1/2 ⁺	
6222.7	(7/2 ⁻ ,9/2,11/2 ⁻)	1132 [†] 1	<25 [†]	5090.2	(11/2 ⁻)	
		1260 [†] 1	100 [†] 25	4962.8	(9/2 ⁻)	
		1729 [†] 1	100 [†] 25	4494.1	(7/2 ⁻)	
7526.9	(1/2 ⁻)	7526 2		0	1/2 ⁺	

[†] From ²⁰⁸Pb(³⁶S,³⁵P γ).

[‡] From ⁹Be(³⁶S,³⁵P γ).

Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

Adopted Levels, GammasLevel Scheme

Intensities: Relative photon branching from each level

